# BarCo Corporation Supply Chain Optimisation

Individual Video Presentation — Speech Script

49680 Value Chain Engineering Systems · Autumn 2026 · UTS FEIT · Group Z

**Estimated speaking time: 5–6 minutes** (~700 words at 120–140 wpm)

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## SLIDE 1 — TITLE (~30 seconds)

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Good morning, everyone. My name is Botao Huo, student ID 25104579, from Group Z. Today I'm presenting our group's analysis on how BarCo Corporation can best manage a natural gas **curtailment** [k■■r-■te■l-m■nt] crisis using a technique called **Linear Programming** [■l■n-i-■r ■pro■-■ræm-■■].

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## SLIDE 2 — OVERVIEW (~15 seconds)

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I'll cover four areas: first, the business problem; second, our analysis approach; third, the key results; and finally, our strategic recommendations.

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## SLIDE 3 — BUSINESS PROBLEM (~45 seconds)

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BarCo Corporation operates eight chemical product lines across three manufacturing plants. The company's energy supplier, MaxEnergy, has warned of natural gas shortages caused by an extreme heat wave — electrical power plants are running at full capacity, and natural gas remains the dominant boiler fuel.

MaxEnergy plans rolling **brownouts** [■bra■n-a■ts] — that means temporary, periodic reductions in gas supply — of between 20 and 40 percent. The Federal Commission has established priority rules: residential heating comes first; industrial raw material users come second; and industrial boiler fuel users come third. Most of BarCo's operations fall into the second and third categories, so BarCo must absorb a large portion of the cut itself.

The key question is: which products should BarCo reduce, and by how much, to protect as much profit as possible?

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## SLIDE 4 — FINANCIAL RISK (~30 seconds)

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At the current baseline, BarCo earns $611,062 per day. Without a smart plan, a 40 percent curtailment could cut profit by about 33 percent. However, with our LP approach, profit drops less than proportionally to gas — a key advantage worth approximately $16 million per year over doing nothing.

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## SLIDE 5 — ANALYSIS APPROACH (~40 seconds)

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To answer this problem, we used Linear Programming — a mathematical **optimisation** [■■p-t■-ma■-■ze■-■■n] technique designed exactly for this type of situation: maximise profit subject to **constraints** [k■n-■stre■nts].

We built two models. The first is a production LP that maximises daily profit under a gas budget. Each product has a known profit per ton, gas consumption per ton, and a daily capacity ceiling. The second is a distribution LP that minimises gas transport costs across the pipeline network connecting MaxEnergy's three supply nodes to BarCo's three plants. Both models were solved using the **Simplex** [■s■m-pleks] **algorithm** [■æl-■■-r■ð-■m].

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## SLIDE 6 — ASSUMPTIONS (~20 seconds)

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Key assumptions include: profits and gas consumption rates are constant per ton — we call this linearity; production volumes can be any real number — divisibility; all cost values are known with certainty; and curtailment applies equally to all three supply nodes. **Sensitivity** [■sen-s■-■t■v-■-ti] analysis later tests how robust our results are when these assumptions are relaxed.

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## SLIDE 7 — PRODUCT ECONOMICS (~35 seconds)

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This chart shows the key insight: when gas is scarce, what matters is not just profit per ton, but profit per unit of gas consumed. **Ammonium** [■-■mo■-ni-■m] **phosphate** [■f■s-fe■t] earns $14 per thousand cubic feet of gas — the highest in our portfolio. **Urea** [j■■-■ri■-■] comes second at $10.36. At the bottom sits **hydrofluoric** [■ha■-dr■-■fl■■r-■k] acid at just $5.59. The LP will protect high-value products first, and cut low-value ones first. This is not a guess — it is the mathematically exact LP solution.

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## SLIDE 8 — PRODUCTION RESULTS (~45 seconds)

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This table shows the optimal production mix at each curtailment level. At baseline, we achieve $611,062 per day. Under a 20 percent cut, profit falls to $513,023 — a 16 percent drop despite a 20 percent gas reduction. Under 40 percent curtailment, profit falls to $411,623 — a 32.6 percent drop.

The cut sequence is: hydrofluoric acid first, then ammonia, then **vinyl** [■va■-n■l] **chloride** [■kl■■r-a■d] **monomer** [■m■n-■-m■r], then **caustic** [■k■■s-t■k] soda. Our four protected products — ammonium phosphate, urea, chlorine, and ammonium **nitrate** [■na■-tre■t] — remain at full capacity even under a 50 percent cut. Versus a simple proportional cut, this LP approach saves $44,000 per day — worth $16 million **annualised** [■æn-ju-■-la■zd].

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## SLIDE 9 — DISTRIBUTION RESULTS (~30 seconds)

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For gas distribution, the optimal plan uses five pipeline lanes: I to B, J to A, J to B, K to B, and K to C. The total transport cost at baseline is $113,952 per day. Remarkably, this structural pattern does not change under any level of curtailment — only the volumes shrink proportionally. This means BarCo can pre-negotiate the same five contracts and simply adjust volumes, eliminating the need to re-tender with each brownout.

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## SLIDE 10 — SENSITIVITY ANALYSIS (~30 seconds)

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Our **sensitivity** [■sen-s■-■t■v-■-ti] analysis shows that every extra 1,000 cubic feet of gas above the curtailed budget is worth between $5.59 and $6.25 in daily profit. This is called the shadow price. It gives management a clear maximum price to pay for emergency gas procurement or fuel-switching. On profit **coefficients** [■ko■-■-■f■■-■nts]: our four protected products can absorb price changes of plus or minus 20 percent without any change to the optimal mix. Marginal products are more sensitive — monitor them closely during volatile market periods.

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## SLIDE 11 — ENVIRONMENTAL SCENARIO (~20 seconds)

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On sustainability: the same LP model, extended with one extra constraint, can minimise CO■ emissions. If BarCo accepts a 10 percent reduction in profit, it achieves a 17 percent reduction in CO■. The break-even carbon price is approximately $80,000 per tonne — well above Australia's current Safeguard Mechanism price of around $30. Today, profit maximisation remains optimal, but this model positions BarCo ahead of future carbon regulation.

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### SLIDE 12 — RECOMMENDATIONS (~50 seconds)

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Based on our analysis, we make three recommendations.

Recommendation One: Adopt the LP-derived production priority rule as a written policy. Always run ammonium phosphate, urea, chlorine, and ammonium nitrate at full capacity. The official cut sequence is: hydrofluoric acid first, then ammonia, then vinyl chloride monomer, then caustic soda. This rule alone saves approximately $16 million per year compared to a naive approach. The shadow price also tells management the exact price ceiling for emergency gas procurement.

Recommendation Two: Pre-commit to the five-lane distribution structure with volume-flexible contracts. There is no need to re-tender each cycle — the optimal routing is structurally stable. Also renegotiate the currently unused and expensive lanes — I to C and K to A — to retain optionality if curtailment patterns change.

Recommendation Three: Run the CO■-aware LP model in parallel every planning cycle. Report the trade-off to the board. Monitor Australia's Safeguard Mechanism carbon price, and activate the emissions-aware production plan as soon as carbon prices become economically significant.

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### SLIDE 13 — CONCLUSION (~20 seconds)

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In summary: Linear Programming transforms BarCo's curtailment challenge from a reactive, high-pressure guessing game into a proactive, quantified policy. With these three recommendations, BarCo is prepared not just for the 2026 brownouts, but for a more constrained and carbon-aware future ahead.

Thank you very much for listening. I'm happy to take any questions.

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## Phonetic Guide to Difficult Terms

*Terms in bold appear in the script with their phonetic notation. Use this page as a quick reference before recording.*

| Term | Phonetic Notation |
| --- | --- |
| **curtailment** | [k■■r-■te■l-m■nt] |
| **linear programming** | [■l■n-i-■r ■pro■-■ræm-■■] |

| | |
|---|---|
| **optimisation** | [■■p-t■-ma■-■ze■-■■n] |
| **algorithm** | [■æl-■■-r■ð-■m] |
| **simplex** | [■s■m-pleks] |
| **constraint** | [k■n-■stre■nt] |
| **coefficient** | [■ko■-■-■f■■-■nt] |
| **sensitivity** | [■sen-s■-■t■v-■-ti] |
| **annualised** | [■æn-ju-■-la■zd] |
| **ammonium** | [■-■mo■-ni-■m] |
| **phosphate** | [■f■s-fe■t] |
| **urea** | [j■■-■ri■-■] |
| **nitrate** | [■na■-tre■t] |
| **hydrofluoric acid** | [■ha■-dr■-■fl■■r-■k ■æs-■d] |
| **vinyl** | [■va■-n■l] |
| **chloride** | [■kl■■r-a■d] |
| **monomer** | [■m■n-■-m■r] |
| **caustic** | [■k■■s-t■k] |
| **brownout** | [■bra■n-a■t] |
| **formulation** | [■f■■r-mj■-■le■-■■n] |
| **scenario** | [s■-■nær-i-o■] |
| **prioritisation** | [pra■-■■r-■-t■-■ze■-■■n] |

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curtailment [k■■r-■te■l-m■nt]

linear programming [■l■n-i-■r ■pro■-■ræm-■■]

optimisation [■■p-t■-ma■-■ze■-■■n]

algorithm [■æl-■■-r■ð-■m]

simplex [■s■m-pleks]

constraint [k■n-■stre■nt]

coefficient [■ko■-■-■f■■-■nt]

sensitivity [■sen-s■-■t■v-■-ti]

annualised [■æn-ju-■-la■zd]

ammonium [■-■mo■-ni-■m]

phosphate [■f■s-fe■t]

urea [j■■-■ri■-■]

nitrate [■na■-tre■t]

hydrofluoric acid [■ha■-dr■-■fl■■r-■k ■æs-■d]

vinyl [■va■-n■l]

chloride [■kl■■r-a■d]

monomer [■m■n-■-m■r]

caustic [■k■■s-t■k]

brownout [■bra■n-a■t]

formulation [■f■■r-mj■-■le■-■■n]

scenario [s■-■nær-i-o■]

prioritisation [pra■-■■r-■-t■-■ze■-■■n]